

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #14 - Healthcare & Telemedicine - Corporate Mental Wellness & Counseling Platform

Scenario: An anonymous counselling portal for a company. Employees can check their stress levels, book confidential sessions with a licensed therapist and follow their own wellness indicators, and at no point does the employer learn who used it.

Stated assumptions

Assessment instrument. The questionnaire is ten items, each answered on a five-point frequency scale (Never 0, Rarely 1, Sometimes 2, Often 3, Very Often 4), giving a total out of 40. Bands are low (0-10), moderate (11-20), high (21-30) and severe (31-40). This is a wellness self-assessment for triage and self-awareness. It is not a diagnostic instrument and the platform does not present it as one.

Two separate identities. The employee has a corporate identity (employee ID and corporate email) used once, to prove they are entitled to the benefit. Everything on the counselling side is keyed on a pseudo-ID such as WEL-7F42A91C, generated at random rather than derived from the corporate identity, so it cannot be reversed.

Where the link is kept. The pseudo-ID to employee mapping lives in an identity store separate from counselling records. Booking, assessment and notes data hold the pseudo-ID and nothing else, so a leak of the counselling database does not identify anybody.

Out of scope for Lab 1. Session delivery (the video or chat call itself), therapist licensing checks and payment or billing are outside this iteration.

Deliverable 1: Requirements table

Req ID	Type	Description	Priority	Acceptance criteria	Rationale
FR-001	Functional	The system shall verify an employee against the corporate directory once at registration, then issue a randomly generated pseudo-ID, and shall use only that pseudo-ID in every record and screen available to a therapist, so that no corporate email address, employee ID or name is disclosed.	High	Pass: after an employee books a session, the therapist dashboard and the stored booking record show only the pseudo-ID in the format WEL-XXXXXXX, and the identity mapping is unreadable from the counselling database. Fail: any booking metadata field contains the employee's email, employee ID or full name, or the pseudo-ID can be derived from the corporate identity.	The whole service depends on this. An employee who thinks HR can see their name simply will not book.
FR-002	Functional	The system shall let an employee complete a ten-item stress self-assessment, each item answered on a five-point frequency scale, and shall return the total score out of 40 with its band (low, moderate, high or severe) once the final item is answered.	High	Pass: submitting a fully answered questionnaire returns a score between 0 and 40 and the band matching the published ranges, within 2 seconds, labelled as a wellness check rather than a diagnosis. Fail: no result is returned, the band contradicts the score, or a partly answered form is scored.	Gives the employee and the therapist the same starting point. The score is also what FR-005 checks against.
FR-003	Functional	The system shall allow an employee to book a counselling session only in a slot the therapist has published as free, and shall mark that slot unavailable in the same transaction.	High	Pass: when two employees confirm the same slot at the same moment, one booking is confirmed and the other receives a 'slot no longer available' message. Fail: both bookings are confirmed, or a slot outside the therapist's published availability is accepted.	Stops double booking. A therapist who finds two people waiting for the same slot has to turn one of them away.
FR-004	Functional	The system shall show an employee a trend of their own wellness indicators (stress score,	Medium	Pass: an account with six or more assessments plots all six in date order; an account with fewer plots only the assessments	A trend is easier to read than a list of old scores, and it gives the employee

		sleep hours and sessions attended) across their six most recent assessments.		that exist. Fail: the chart shows another account's data, or fills dates that have no assessment with estimated values.	a reason to keep taking the assessment.
FR-005	Functional	The system shall, whenever an assessment falls in the severe band, display regional crisis helpline details on the results screen and offer the earliest available priority appointment slot.	High	Pass: a submission scoring in the severe band shows the helpline panel and lists at least one priority slot starting within the next 24 hours. Fail: a severe score is presented as an ordinary result, with no helpline and no priority slot.	A severe score needs attention the same day, not after a two-day wait in the normal booking queue.
NFR-001	Nonfunctional (Performance & Security)	The appointment scheduler shall convert every slot to the viewer's local timezone correctly, including across daylight saving transitions, and shall return a slot list in under 2 seconds at peak load.	High	Pass: a benchmark of 500 concurrent users across IST, GMT and PST keeps 95th percentile slot-list latency below 2 seconds, and a booking made on a daylight saving changeover date shows the same absolute time to both parties. Fail: the two views disagree on the time, or 95th percentile latency exceeds 2 seconds.	Employees and therapists sit in different countries and book across timezones daily. An hour's error means somebody misses the session.
NFR-002	Nonfunctional (Security & Privacy)	The system shall hold the pseudo-ID to employee mapping in a store separate from all counselling data, shall encrypt that mapping, session notes and assessment answers with AES-256 at rest and TLS 1.3 in transit, and shall release an aggregate wellness report to a Wellness Program Administrator only where the reported group contains at least ten employees.	High	Pass: a dump of the counselling database yields no readable notes and no corporate identifier, and a report requested for a department of nine employees is refused with a 'group too small' message. Fail: plaintext notes appear in storage or logs, the mapping is readable from counselling data, or a report is released for a group smaller than ten.	Session notes are clinical records, and the mapping is the one table that would undo every pseudo-ID at once. Reports on very small teams would also let HR work out who has been using the service.

Actors

Actor	Kind	Role in the system
Corporate Employee	Primary	Takes assessments, books and attends sessions, tracks their own indicators. Known to the counselling side of the system only by a pseudo-ID.
Licensed Therapist	Primary	Publishes availability, conducts sessions and records notes. Sees the pseudo-ID, never the employee's corporate identity.
Wellness Program Administrator	Primary	HR-side owner of the programme. Reads aggregate uptake and wellness reports only, subject to the group size floor in NFR-002.
Corporate Identity Provider	Supporting	The employer's existing directory or SSO. Confirms once that the person is a current employee, and is never told what they do afterwards.
Crisis Support Service	Supporting	External helpline the platform hands off to when an assessment scores in the severe band.
Notification Service	Supporting	External email and push provider that delivers booking confirmations and reminders.

Use cases and traceability

Use case	Title	Actor / relationship	Traces to
UC-01	Create Anonymous Wellness Account	Corporate Employee	FR-001

UC-02	Take Stress Self-Assessment	Corporate Employee	FR-002
UC-03	Book Counseling Session	Corporate Employee, Licensed Therapist	FR-003
UC-04	View Wellness Trend Dashboard	Corporate Employee	FR-004
UC-05	Manage Availability Calendar	Licensed Therapist	FR-003
UC-06	Record Encrypted Session Notes	Licensed Therapist	NFR-002
UC-07	View Aggregate Wellness Report	Wellness Program Administrator	NFR-002
UC-08	Generate Pseudo-ID	included by UC-01 and UC-03	FR-001
UC-09	Escalate to Crisis Support	extends UC-02	FR-005
UC-10	Verify Slot Availability	included by UC-03	FR-003
UC-11	Send Confirmation Notification	included by UC-03	FR-003, NFR-001
UC-12	Reschedule or Cancel Session	extends UC-03	FR-003
UC-13	Verify Employment Status	included by UC-01	FR-001

Include and extend relationships

Relationship	Why it is modelled this way
UC-01 «include» UC-13	Registration always checks the person against the corporate directory first. An account that skipped the check would make the pseudo-ID meaningless.
UC-01 «include» UC-08	Account creation always generates a pseudo-ID; there is no account without one.
UC-03 «include» UC-08	Every booking is written against a pseudo-ID, so the identifier is resolved on each booking as well.
UC-03 «include» UC-10	A booking always checks the slot is still free before it is committed.
UC-03 «include» UC-11	A confirmed booking always sends a confirmation to both parties.
UC-09 «extend» UC-02	Crisis escalation runs only when the assessment scores in the severe band.
UC-12 «extend» UC-03	Rescheduling or cancelling happens only if the employee or therapist later changes the booking.